

PRACTICE QUIZ ON BIOENERGETICS

Instructions: You may prefer to print out the quiz and mark it. After you have finished marking the quiz, you can grade each question by clicking on a quiz answer. When you click on a quiz answer, you will be given feedback indicating whether your answer is correct. *When you get a feedback response, click on the back arrow to return to the quiz.* After scoring your quiz, you should have an idea if you need to study the material more. Remember to post questions about things you don't understand in the Class Forum.

1. The reaction $\text{ATP} \rightarrow \text{ADP} + \text{phosphate} + \text{energy}$ indicates that
 - a. energy has been trapped
 - b. energy has been released
 - c. ATP has been formed
 - d. none of these
2. One molecule of which of the following contains the greatest amount of energy?
 - a. glucose
 - b. ATP
 - c. pyruvic acid
 - d. lactic acid
3. Oxidation can be described as
 - a. a loss of H
 - b. a gain of energy
 - c. gain of O
 - d. (a) and (c) are correct
 - e. (a), (b), and (c) are correct
4. Which releases the greatest amount of usable energy per molecule of glucose broken down?
AR = Aerobic Respiration; F = Fermentation; G= Glycolysis
 - a. AR in amoeba
 - b. F in yeast
 - c. G in a human liver cell
 - d. F in a human muscle cell
5. Fermentation is necessary for the recycling of
 - a. ATP
 - b. NAD⁺
 - c. energy
 - d. lactic acid
6. Which of the following is NOT an end product in aerobic respiration?
 - a. H₂O
 - b. CO₂
 - c. ATP
 - d. lactic acid
7. The Citric Acid Cycle in eukaryotic cells takes place within the _____ of the cell.
 - a. mitochondria
 - b. nucleus
 - c. cell membrane
 - d. cytoplasm
8. Which of the following is an exergonic reaction?
 - a. oxidation of glucose to CO₂
 - b. breakdown of ATP into ADP+ Pi
 - c. both (a) and (b)
 - d. neither (a) nor (b)

9. For each glucose molecule, glycolysis produces

- a. 36 ATPs
b. citric acid & 36 ATPs
c. ethyl alcohol, CO₂, & ATP
d. 2 pyruvates, 2 ATP, & 2 NADH + 2 H⁺

10. The greatest amount of ATP is formed during

- a. Glycolysis b. Fermentation c. Citric Acid Cycle d. Chemiosmotic Phosphorylation

11. The bubbles in champagne, which is a fermentation product, are due to

- a. oxygen b. carbon dioxide c. ATP d. glucose

12-14. Match the basic processes of photosynthesis listed below with questions

12 - 14. Be sure to answer all three questions before checking your answers.

- a. light dependent reaction
- b. light independent reaction
- c. both light dependent and light independent reactions

12. reduction of CO₂ to glucose

13. light-energized electrons

14. splitting of water

15. In the reaction $\text{NADH}_2 + \text{FAD} \rightarrow \text{NAD}^+ + \text{FADH}_2$, which reactant is oxidized as the reaction proceeds to the right?

- a. NADH_2 b. FAD c. NAD^+ d. FADH_2

16-21. Explain what is accomplished in each of the following processes:

16. cellular respiration under aerobic conditions ANSWER

17. glycolysis ANSWER

18. alcoholic fermentation ANSWER

19. lactic acid fermentation ANSWER

20. citric acid cycle ANSWER

21. chemiosmotic phosphorylation ANSWER